

Exercise 1

1

Sort the arcs by increasing cost

(1,2) → added

(6,7) → added

(1,5) → added

(1,3) → added

(3,5) → discarded

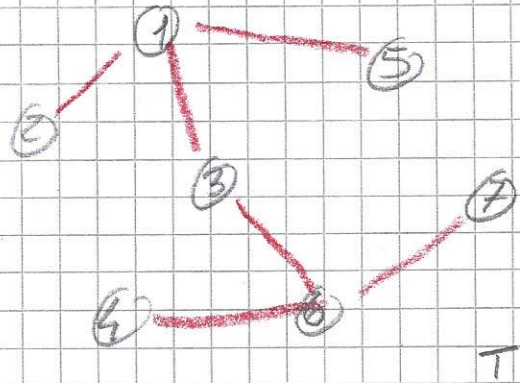
(3,6) → added

(2,3) → discarded

(5,7) → discarded

(5,6) → discarded

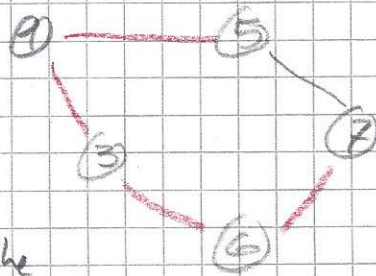
(4,6) → added



Stop since the solution is a spanning tree T

Arc (5,7) $\notin T$

if we add it to the solution we have a circuit



the cost of (5,7)

must remain $\leq$ than the

costs of the other arcs of the circuit

$$c_{57} \leq \max \{4, 5, 3, 7\} = 7$$

Arc (3,6) $\in T$

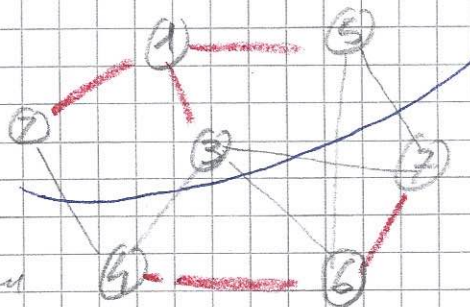
if we remove (3,6) from T

we have a cut

The cost of (3,6) must remain

$\leq$ than the cost of the other arcs traversing the cut

$$c_{36} \leq \min \{16, 15, 10, 13, 2\} = 2$$



Exercise 2

(2)

$x = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ is not feasible since it violates constraint 1

$$x_1 + 3x_2 \leq -3$$

$$0 \not\leq -3$$

The auxiliary problem is the following:

$$\begin{aligned} \text{AP: max} \quad & -u_1 \\ & x_1 + 3x_2 - u_1 \leq -3 \\ & -2x_1 + x_2 \leq 0 \\ & x_1 - 4x_2 \leq 12 \\ & -u_1 \leq 0 \end{aligned}$$

An obvious feasible solution for AP is $\begin{bmatrix} x_1 \\ x_2 \\ u_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}$

The active constraints are

$$I = \{1, 2\}$$

$$A_I = \begin{bmatrix} 1 & 3 & -1 \\ -2 & 1 & 0 \end{bmatrix}$$

The conditions for the existence of a growing feasible directions are

$$\begin{cases} c \notin \mathcal{D} \\ A_I \xi \leq 0 \end{cases}$$

$$\begin{cases} -\xi_3 > 0 \\ \xi_1 + 3\xi_2 - \xi_3 \leq 0 \\ -2\xi_1 + \xi_2 \leq 0 \end{cases}$$

Exercice 3

(3)

$$\min 5x_1 - x_2$$

$$3x_1 + 3x_2 + x_3 = 13$$

$$-8x_1 + 3x_2 + x_4 = 8$$

$$x_1, x_2, x_3, x_4 \geq 0 \text{ integers}$$

$$B = \{2, 3\}$$

$$A_B = \begin{bmatrix} 3 & 1 \\ 3 & 0 \end{bmatrix}$$

$$A_N = \begin{bmatrix} 3 & 0 \\ -8 & 1 \end{bmatrix}$$

$$A_B^{-1} = \begin{bmatrix} 0 & 1/3 \\ 1 & -1 \end{bmatrix}$$

$$\bar{x}_B = A_B^{-1} b = \begin{bmatrix} 0 & 1/3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 13 \\ 8 \end{bmatrix} = \begin{bmatrix} 8/3 \\ 5 \end{bmatrix}$$

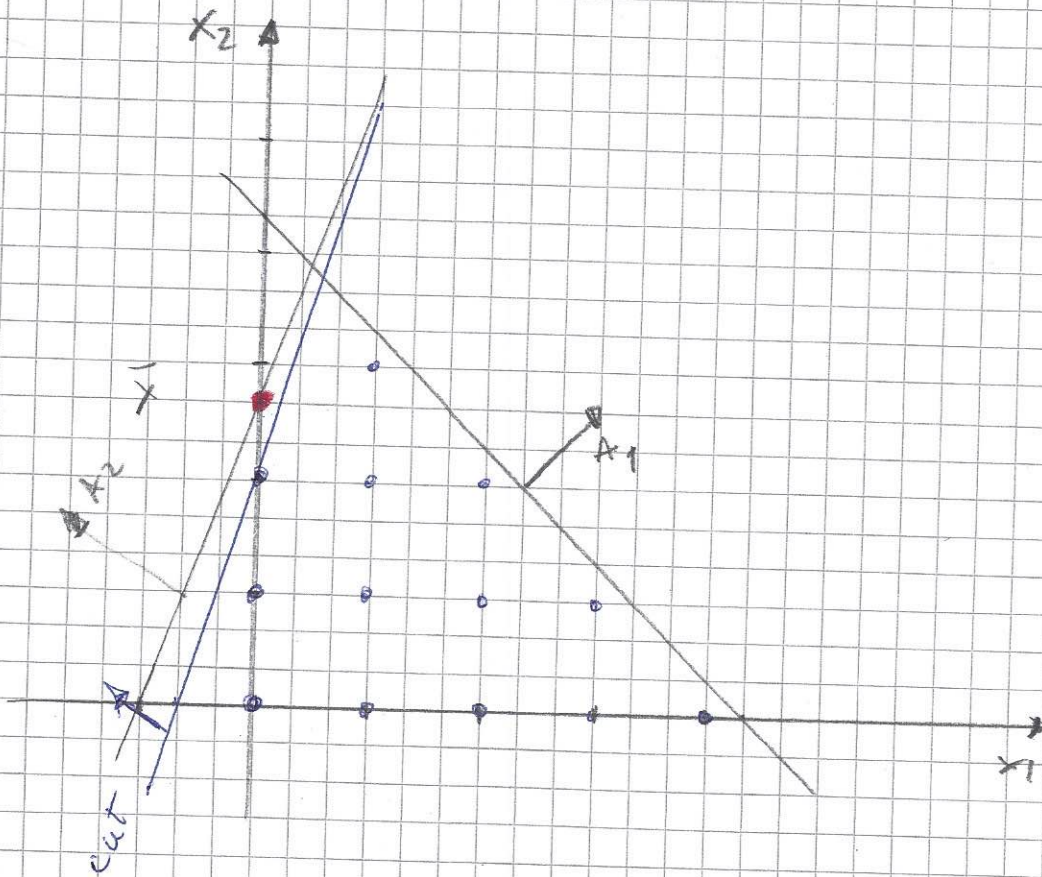
$\bar{x}_2$ is fractional

$$A_B^{-1} A_N = \begin{bmatrix} 0 & 1/3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ -8 & 1 \end{bmatrix} = \begin{bmatrix} -8/3 & 1/3 \\ 11 & -1 \end{bmatrix} \begin{matrix} x_1 \\ x_4 \end{matrix}$$

Gomory cut

$$x_2 + \left\lfloor -\frac{8}{3} \right\rfloor x_1 + \left\lfloor \frac{1}{3} \right\rfloor x_4 \leq \left\lfloor \frac{8}{3} \right\rfloor$$

$$x_2 - 3x_1 \leq 2$$



Exercise 4

④

P' is a relaxation of P if $F \subseteq F'$

$$c x' \geq c x^* \quad \text{since } x^* \in F'$$

but F' may include also other solutions
having a greater value

if P' has no feasible solution then $F' = \emptyset$

thus $F = \emptyset$ too being a subset of F'

(5)

Write the definition in the AMPL modeling language of the following:

- a parameter n
- a set I of all the integers between 1 and n .
- a set of positive variables x_i with $i \in \{I \cup 0\}$.
- a set of parameters b_i with $i \in I$.

Then using the defined variables and parameters write in the AMPL modeling language the following constraint:

$$x_{i-1} \leq b_i x_i$$

$$\forall i \in I : i \leq 5$$

0.1 solution

```
param n;  
set I;  
set I_bar := {I union {0}};  
var x{I_bar} >= 0;  
param b{I};  
  
s.t. c1{i in I : i <= 5}:  
    x[i-1] <= b[i] * x[i];
```

ANSWER IN THE SPACE BELOW

Exercise 5

⑥

Sets

$O = \{1 \dots n\}$ origins

$D = \{1 \dots m\}$ destinations

Parameters

a_i $i \in O$ workers in i

b_j $j \in D$ jobs in j

c_{ij} $i \in O, j \in D$ cost of moving i to j

s_i $i \in O$ cost of firing one worker in i

variables

x_{ij} = number of workers moved from i to j

y_i = number of fired workers in i

objective function

$$\min \sum_{i \in O} \sum_{j \in D} c_{ij} x_{ij} + \sum_{i \in O} s_i y_i$$

constraints

$$\sum_{j \in D} x_{ij} + y_i = a_i \quad i \in O$$

all workers of i are moved or fired

$$\sum_{i \in O} x_{ij} \leq b_j \quad j \in D$$

capacity constraint in j

$$x_{ij} \geq 0 \quad i \in O, j \in D$$

$$y_i \geq 0 \quad i \in O$$

Variant

parameters F_i $i \in O$ limit in the number of firings

P_i $i \in O$ penalty for exceeding the limit

variables $z_i = \begin{cases} 1 & \text{if the limit is exceeded} \\ 0 & \end{cases}$

obj function $\min \sum_{i \in O} \sum_{j \in D} c_{ij} x_{ij} + \sum_{i \in O} s_i y_i + \sum_{i \in O} P_i z_i$

constraints $y_i \leq F_i + a_i z_i \quad i \in O$

Exercise 6

7

$$E = R$$

$$F \subseteq 2^R$$

family of subsets of R such that
if j and h belong to the subset
 $S_j \cap S_h = \emptyset$

Let C be the subset of recipes to be considered
(at the beginning $C = R$)

$$\text{Best}(C) = \underset{\text{minimum cardinality, recipe}}{\operatorname{argmin}} \{ |S_j| \mid j \in C \}$$

Let X be the partial solution

$$\text{Ind}(X, j) = \begin{cases} \text{if } \forall h \in X \quad S_h \cap S_j = \emptyset & \text{then true} \\ & \text{else false} \end{cases}$$

variant

$$\text{Best}(C) = \underset{\text{minimum cardinality, recipe}}{\operatorname{argmax}} \left\{ \frac{z_j}{|S_j|} \mid j \in C \right\}$$